Brambleloop: faithful crochet lifestyle imagery

Independent technical research | 19 October 2026

Decision in one minute

There are credible newer tools worth testing. My first implementation candidate is Ideogram 4.5 Precise Edit feeding Brambleloop's existing protected-product compositor. My quality challenger is GPT Image 2.5 Sunburst. This is a ranked engineering shortlist, not a claim of measured crochet superiority. No images were generated in this research and all acceptance rates remain unknown.

The strongest solution is a hybrid: a verified, sufficiently realistic product asset + a carefully checked alpha mask + scene generation + deterministic foreground preservation + independent Product Truth verification. This need not mean manufacturing photographs are the only route. A genuinely validated photorealistic render can also be an input. A diagram or certified informational gallery, however, is not automatically that render.

A single generative model cannot presently be shown to guarantee that a newly synthesized object exactly implements an arbitrary crochet pattern. A narrower guarantee is feasible: preserve an approved foreground while changing its surroundings. Exact foreground retention does not prove the reference was correct, and does not provide a new camera angle, drape or hidden backside. Separate those questions instead of buying more attractive redraws.

Best first experiment: a staged, maximum CA$30 paid-usage ceiling, requiring owner approval, with a cheap early stop if references are unsuitable. Test the same three products across five arms, score every attempt, and measure cost per accepted image. Do not spend until the references and price reservations are approved.

Research boundary: current public provider documentation, official Etsy policy and read-only repository inspection. Provider capabilities and terms are documented facts; fit rankings and architecture are recommendations. No paid account entitlement, latency, output quality, physical sample, production deployment or acceptance gate has been independently proved by this desk research.

Answers to the key questions

- Best reference-fidelity candidate: Ideogram 4.5 Precise Edit for background-only edits, with our own protected-layer copy-back. This is a narrow, testable preservation approach, not a new-pose guarantee.

- Best photorealistic scene quality candidate: GPT Image 2.5 Sunburst, provisionally. FLUX 3 Image and Gemini 3 Pro Image are strong comparison candidates. No crochet-specific head-to-head evidence establishes a winner.

- Best practical automated editor: Ideogram 4.5 Precise Edit, because of masked local editing, explicit context-region controls, pricing and asynchronous operation. Recraft V3 is a useful low-cost masked baseline.

- Best specialist workflow to investigate: Photoroom Studio with supplied alpha, original placement and final deterministic copy-back; Adobe Precise Composite is a higher-touch alternative with quote-dependent pricing.

- Best hybrid: keep the current compositor and certification pipeline, add one provider adapter, enforce artifact provenance, and admit only independently qualified whole-product references.

- CA$4.76: plausible as a small generation-only allowance, not a credible promise to finish eight truth-certified lifestyles. Reference creation, matting, failed attempts, evaluation and deployment are outside that figure.

Evidence and currency convention

All CAD conversions use 1 USD = CA$1.4154, Bank of Canada observation 16 October 2026, the latest published observation available on the research morning. These are reference conversions before tax, card spread, minimum purchases and labor. Do not label this an observed 19 October FX rate. [S1]

Documented API availability does not prove account entitlement. ChatGPT/Claude subscriptions are not assumed to fund APIs.

Top five models, ranked for this specific problem

Ranking prioritizes controllable preservation before beauty. A lower-ranked model could win a blind realism test. Specialized compositing services are evaluated separately because their underlying generative model may not be disclosed.

Rank / exact model	Why it is here	Principal limitation	API cost in CAD
1. Ideogram 4.5 Precise Edit	Best documented local-edit fit; mask and context window; retain source layer	Copy wording is qualified; must independently test unchanged area and alpha edges	Medium $0.0849; high $0.3114 per 1K/2K result
2. Recraft V3 masked inpaint / generateBackground	Explicit protected-mask workflow, lossless output; cheap baseline	V4.1 is newer for generation, but V3 remains the documented editing model; texture synthesis unproved	$0.0566 per raster edit
3. gpt-image-2.5-sunburst-2026-09-08	Premium detailed editing and scene candidate; snapshot can be pinned	Mask is guidance, not a hard pixel lock; token cost varies	Illustrative $0.1960-$0.3659, not a fixed quote
4. gemini-3-pro-image	Premium multi-reference precision candidate and up to 4K	Semantic edits can change product; no immutable-foreground contract	2K illustrative lower bound $0.1932; 4K output alone $0.3397
5. FLUX 3 Image, /v1/flux-3-image	Ten reference inputs, layout control, high-resolution scene potential	Only latest version documented; exact tier price not conclusively verified; no raster protection mask established	Quote required; do not misapply FLUX.2 pricing

Sources: Ideogram [S2-S4], Recraft [S5-S7], OpenAI [S8-S11], Google [S12-S14], BFL [S15-S17].

Why Recraft outranks a newer full-image generator here: the task is retaining actual stitches. A local mask contract and a deterministic baseline are more directly useful than a generic photorealism ranking. This does not assert that Recraft produces better-looking rooms.

Efficiency challenger: gemini-nano-banana-2.1, GA 6 October 2026, is especially worth testing at a 2K request lower bound of CA$0.0748. It may be preferable to Pro for operating cost if measured acceptance is comparable. [S12-S14]

Capability matrix: the serious finalists

D = documented operation; U = untested for Brambleloop. All five have unknown crochet acceptance rates.

Capability	Ideogram 4.5	Recraft V3	Sunburst 2.5	Gemini 3 Pro	FLUX 3 Image
Image-to-image / editing	D	D	D	D	D
Reference fidelity	Local-edit mechanism; U	Mask mechanism; U	Semantic; U	Semantic; U	Semantic; U
Repeated stitch preservation	Protected layer testable	Protected layer testable	U if regenerated	U if regenerated	U if regenerated
Multiple references	3 extra with mask	Styles, not identity lock	Up to 16 inputs	Up to 14 inputs	Up to 10 inputs
Background replacement	Controlled mask	Supplied-mask tool	Guided edit	Conversational edit	Prompt/region edit
Raster mask	Black edit, white keep	White edit, black keep	Mask guidance	No hard lock documented	Raster mask not established
Camera / light / composition	Prompt + edit region	Prompt + mask	Prompt	Prompt + references	Prompt + boxes
Colours / motifs / piece count	Enforce externally	Enforce externally	Enforce externally	Enforce externally	Enforce externally
Photorealism	U on crochet	U on crochet	Premium candidate, U	Premium candidate, U	Premium candidate, U

Capability	Ideogram 4.5	Recraft V3	Sunburst 2.5	Gemini 3 Pro	FLUX 3 Image
Cross-image consistency	Same source layer	Same source layer	References, can drift	References, can drift	References, can drift
Resolution	Source-sized edit with context	Large input accepted; verify output	Up to 8.29MP; >2560x1440 experimental	Up to 4K	Up to 4K / roughly 16MP
Automation	Sync/async, poll/webhook	REST, temporary assets	Images/Responses APIs	Gemini API / batch	Async polling
Repeatability	Seed available; still verify	Log settings; verify	Pin snapshot; no pixel promise	Record returned version	latest can change
Latency / reliability	Measure p50/p95	Measure p50/p95	Measure p50/p95	Measure p50/p95	Measure p50/p95
Commercial / Etsy	Rights subject to terms; Etsy separate	Same	Same	Same	Same
Production integration	Thin adapter + current guards	Thin adapter + current guards	Existing adapter may need new ID	Thin adapter	Thin adapter + price verification

Any advertised subject consistency is weaker than a verified construction-preservation result. No reviewed evidence supplies a crochet-specific, independently repeatable acceptance rate.

Wider search: serious alternatives and exclusions

The search included current models, older still-supported editing tools, and specialist product systems. A vendor's newest generator is not necessarily its strongest background-preservation tool.

Candidate / operation	Verified capability and relevant role	CAD per output / request
Gemini Nano Banana 2.1	Up to 14 refs and 1K/2K/4K; efficient semantic editing	2K output $0.0713; one-ref scenario >=$0.0748
GPT Image 2.5 Flare	Current speed-oriented sibling; same token rate, not necessarily same token consumption	Token-metered; quote actual usage
FLUX.2 Pro / Max	Still documented reference editing; separate from FLUX 3	1MP ref +1MP output $0.0637 / $0.1415
FLUX.2 Klein 4B	Apache-2.0 open weights; multi-reference editing	Self-host compute unmeasured; API example 1MP ref+output $0.0212
Adobe Firefly Image Model 5	Current general image/edit family, native 4MP	Applicable API quote not verified
Adobe Precise Composite	Object + background + placement mask; distinguish from Adaptive Composite	Quote required, not consumer plan credits
Photoroom Studio	v2/edit; supplied alpha and original placement; background generation	Plus list rate $0.1415; billing/minimums need confirmation
Bria Lifestyle high_control	Product alpha, text/image backgrounds, placement controls	$0.0425 lifestyle; cutout/shadow each extra
Seedream 5.0 Pro / Flash	Pro 260628; Flash 260915; reference edits and layer decomposition	Pro $0.0637 at <=2.61MP; Flash $0.0255; extra refs may add cost
Qwen Image 3.0 Pro / 3.0	Hosted 1-3 refs; compatible interface lacks mask	Pro 2K +1ref $0.1104; standard +1ref $0.0467
Qwen-Image-Edit-2511	Verified Apache-2.0 20B open editing weights	No license unit fee; GPU/operations not free
Runway gen4_image / turbo	Reference-image generation; useful background candidate	$0.0708 at720p / $0.1132 at1080p; turbo $0.0283

Candidate / operation	Verified capability and relevant role	CAD per output / request
Stability Ultra / Inpaint	SD3.5-based generation; explicit mask editing	Ultra $0.1132; inpaint $0.0708
Stability Replace Background & Relight	Background plus lighting changes	$0.1132; relight changes product pixels
HunyuanImage-3.0-Instruct	Open editing and up to3 refs	Compute not benchmarked; restrictive license
Tencent ReplaceBackground	Custom alpha-mask input; migration to TokenHub complicates access	Mainland CNY price; no CAD quote validated
Flair API	API offering exists, but public model/mask contract insufficient	API early-access/enterprise pricing unresolved

Prices are not comparable quality tiers. A 720p background draft is not a 2000x2000 listing master. Never upscale generated stitches and call them verified detail. Source coverage: [S18-S31].

Licensing, commercial use and confidentiality

- OpenAI API customer input/output rights and paid Google service output rights are subject to terms, lawful source inputs and review. Paid API data handling differs from free tools. Neither grants proof of physical correctness. [S32-S33]

- Ideogram permits commercial output use; its API agreement includes application attribution requirements and exceptions to the no-training treatment. Check the actual agreement before integration. [S34]

- BFL permits commercial outputs, but API terms grant an input/output improvement/training license. Unreleased pattern imagery requires a conscious data-use decision. [S35]

- Recraft API rights permit commercial use but restrict AI training and removal of provenance. Free Studio rules are not the API contract. [S7]

- Adobe commercial use is supported; indemnification and entitlement are contract-dependent. Photoroom enterprise rights cannot silently be assumed to be self-serve terms. Bria indemnity depends on plan. [S19-S21]

- BytePlus and Alibaba hosted terms address output rights separately from open weights. Qwen Edit-2511 and FLUX.2 Klein4B Apache licenses are specific to those checkpoints. Stability's Community License has revenue conditions; Hunyuan's material/output territorial restrictions require special care. [S22-S24,S27-S30]

All candidates remain subject to Etsy's rules. Commercial-use permission is not permission to misrepresent a finished crochet object.

The architecture that actually preserves the product

Lane 1: exact source-preserving lifestyle composition

1. Approve a whole-product reference against the correct version of the actual pattern. Confirm geometry, stitch families, motif topology, borders, joins, dimensions, colour ordering and set membership. A hash certifies identity, not correctness.

2. Preserve an immutable high-resolution master. Prefer owned sample photos or a separately validated photorealistic render. Add front/back and macro views where needed. Keep actual-photo, validated-render and conceptual-illustration provenance distinct.

3. Create a reviewed alpha matte, including yarn fuzz, lace holes, scallops and fringe. Store opaque core, soft edge and true holes separately. Automatic segmentation must not silently fill crochet holes.

4. Choose a room/scene consistent with the existing product pose, light and camera. Generate or source the environment around it. For baskets and coasters, compose separately photographed pieces with stable component IDs and correct count.

5. Keep contact shadows on a separate layer behind the product. Avoid global relighting, generative upscaling or beautification over the finished foreground. Restore the approved foreground deterministically as the last rendering step.

6. Compare the output against the approved compositing baseline; certify the exact final asset hash and route it through the existing Product Truth, listing and preview gates.

For opaque pixels at identical coordinates and colour space, equality can be exact. Soft alpha pixels are mixtures with the new background, so compare the output to the approved alpha equation, not blindly to the old RGB image. Copying every nonzero-alpha source RGB pixel can preserve an old-background halo. This is a matting-quality issue to test; it is not a finding that the repository currently has such a visible defect.

If rotation, perspective or scaling is allowed, compare against one frozen deterministic transformed baseline and record the transformation. That is reproducible preservation of reference content, not literal equality to original pixel coordinates. Do not introduce transforms the current acceptance gates forbid.

Lane 2: changed pose or new drape

A flat blanket cannot become a truth-guaranteed sofa drape by copying its pixels. Either photograph the required pose or build a verified digital twin with actual stitch topology, openings, front/back appearance, edging, dimensions, seams and material behaviour. Preserve motif-to-surface correspondence in rendering. Compare the twin against withheld real views before accepting it.

Structured pattern data is valuable, but it is not automatically a yarn-level renderable mesh. A texture on a smooth rectangle misses holes, thickness, scallops and self-occlusion. Generic single-image-to-3D also invents unseen geometry. This is a larger engineering route; do not make it the default eight-product launch rescue unless reusable infrastructure already exists.

Lane 3: concept exploration

Full-object generative editing can explore scenes, framing and attractive compositions. Keep those candidates quarantined until they pass the same construction tests. A labelled conceptual mockup may be useful internally; it is not automatically acceptable for Etsy or sufficient to close F-030/F-254.

Why generic generation likely struggled

- Text prompts under-specify thousands of repeated loops, joins and local colour relationships.

- Reference encoders often retain semantic identity while redrawing repeated microstructure.

- A new drape exposes information missing from a single reference.

- Multi-piece products invite duplication, omission and accidental fusion.

- Small input crops, compression and creative upscaling remove or invent stitch evidence.

- Global similarity and attractive scene scores can overlook one wrong motif or filled opening.

- Certifying a chart or dimensions card proves different things than certifying a photographic-looking whole product.

These are mechanisms and hypotheses, not claims that every one is a confirmed Brambleloop defect.

Options for eight launch products

Cost scope: two desired lifestyle images per product =16 accepted images. For planning only, allow three generation attempts per desired image =48 attempts. This is a workload assumption, not an observed 33% success rate. Reference preparation, human review, storage, tax and any photography are additional unless explicitly stated.

Option A: best possible quality

Use owned, pattern-verified sample photography in the intended pose, or a truly validated photorealistic render, then GPT Image 2.5 Sunburst for scene proposals and Ideogram 4.5 / local compositing for final foreground preservation.

Adobe Precise Composite is an alternative if its contract and harmonization controls fit.

Strength: strongest credible combination of real product evidence and premium creative scene design. Limit: cannot preserve a photo while arbitrarily changing its light or pose; sample production and capture may dominate schedule and cost.

API: OpenAI dated Sunburst snapshot and Ideogram precise-edit endpoint are documented. Adobe API pricing requires a quote. Integration: medium with usable references; high if reference capture or yarn-level modelling is missing.

48 Sunburst requests at the illustrative token scenarios cost CA$9.41-$17.56. If followed by16 Ideogram medium edits, add CA$1.36: total illustrated API spend CA$10.77-$18.92. These are not capped price quotes. Photo/sample costs are unknown and excluded. Can satisfy F-030/F-254 only if actual resulting assets meet their original evidence and quality criteria.

Option B: best practical automated system

Ideogram 4.5 Precise Edit + the existing protected compositor + the current certification and spend controls. Start at medium quality, escalate only after measured improvement. Use explicit/auto context regions and frozen product masks. Keep the source pose.

Strength: targeted operation, clear unit pricing and enforceable pixel preservation downstream. Limit: source realism and correctness remain prerequisites; mask edges and believable contact must be reviewed.

48 medium attempts cost CA$4.08; 48 high attempts cost CA$14.95, before other expenses. One medium attempt for each of16 images would be CA$1.36, but nobody has established that all16 would pass. Integration estimate: 2-4 engineer-days for a narrow adapter, artifact contract and deterministic tests, assuming reusable assets and existing gates work; not an estimate to finish all remaining requirements.

F-030/F-254: conditional, never closed merely by a provider switch or pixel test. Whole-product truth, visual quality and deployment evidence remain separate.

Option C: lowest-cost reliable system

Use the current deterministic compositor with approved reusable scene templates and owned/approved product foregrounds. Incremental image-model fee can be CA$0 when no new scene generation is needed. If scenes are missing, test Recraft V3 supplied-mask background generation, then hard copy-back.

48 Recraft edits cost CA$2.72;16 one-shot background edits cost CA$0.91, excluding reference/mask work. Bria high_control is another affordable specialist at CA$0.0425 per lifestyle attempt, but validate its size/placement behaviour and final copy-back.

Strength: low cost, repeatable, exact control. Limit: narrower composition diversity, possible cutout appearance and no new drapes. Integration low to medium. F-030/F-254 remain open if a flat or schematic reference cannot produce the required lifestyle quality.

Is CA$4.76 realistic?

It buys about56 Ideogram medium outputs, or84 Recraft V3 edits, at the reference FX before tax/fees. It can fund a bounded exploratory batch. It cannot buy a guarantee, and could be consumed by a handful of high-quality token-metered attempts plus review. The historical repository budget is not today's authorization. Prioritize cost per accepted image = total billed spend / accepted outputs, and report zero accepts as undefined/infinite rather than hiding the failures.

Controlled benchmark: affordable, then evidence-driven

Stage 0: no generation spending

Select three real launch designs or equivalent approved references: (A) heavily textured blanket with repeated rows and intricate border, (B) a multi-piece basket/coaster/ornament set with exact count, (C) complex mosaic or asymmetric motif work. Freeze the same whole-product references, macro crops, component list, palette, source dimensions and pattern-version hashes for every arm. Do not use unverified prior generated images as ground truth.

Have a crochet-competent reviewer mark motifs, borders, joins, holes and piece identities. Confirm rights and confidentiality. Produce a manual/deterministic composite baseline. If no sufficiently realistic whole-product reference can be qualified, stop the exact-preservation benchmark and report that blocker; a schematic-to-photo experiment is a different, unproven synthesis task.

Stage 1: five-arm screening, maximum60 outputs

Three products x two scene briefs x two independent attempts x five arms =60 outputs. Arms: Ideogram4.5 medium, RecraftV3, Sunburst, Nano Banana2.1, Photoroom Studio. The two briefs are (1) a simple background change and (2) a difficult setting with border visibility, near-background colour, crochet openings and multiple pieces. Keep pose fixed. Use the same references and target composition; translate only API syntax and mask polarity. Generate at a common target around2K; document native-size differences rather than quietly upscaling product detail.

Record raw provider output and protected-composite derivative from the same request. This separates model behaviour from our preservation mechanism. Each receives its own review result. No sequential editing chain: every attempt starts from the same approved source. Same numerical seeds do not create equivalent randomness across vendors.

Indicative generation floor for12 calls per arm: Ideogram CA$1.02 + Recraft CA$0.68 + Sunburst CA$2.35-$4.39 illustrative + Nano2.1 >=CA$0.90 + Photoroom CA$1.70 = an illustrative minimum subtotal of approximately CA$6.65-$8.69 before thinking, retries, tax or required plan purchase; actual total can be higher. If Photoroom needs an incompatible minimum subscription, do not buy it; substitute Bria after its contract/quote is verified. FLUX3 or Seedream5 can replace a weak arm in a separately budgeted second round, not silently expand the run.

Proposed ceiling and stopping rules

Request, but do not execute yet: CA$30 total paid usage, with reservation before every call, taxes/FX buffer and evaluation calls included. Stage1 reservations up to CA$20; retain CA$10 for approved repeatability/final-resolution checks. Stop before a request if worst-case cost is not bounded, balance is insufficient or identity/masking fails systematically. A required subscription or credit minimum exceeding the ceiling requires new approval. No auto-top-up.

Blind scoring and acceptance

Two reviewers see randomized provider-free outputs: one construction specialist and one commercial-image reviewer. Separate hard fidelity gates from scene quality. Adjudicate disagreements. Retain rejected outputs, all usage records, retries, elapsed times and manual correction minutes.

Hard gates: exact set count; correct motif count/adjacency; no invented seams or border changes; approved dimensions/proportions; no missing fringe or filled lace holes; matching product version; complete provenance. For protected opaque pixels, maximum channel difference is zero against the approved baseline. For soft edges, verify the alpha-composite baseline and visually inspect halos at200-400%. JPEG delivery derivatives need a separate export check; test identity on the lossless master.

Score realism, believable support/shadows, perspective, composition, texture visibility, thumbnail usefulness and commercial appeal only after fidelity passes. A beautiful wrong object is a rejection. Use AI vision as a second opinion, never the only proof of stitch correctness. SSIM, LPIPS and embeddings can triage; they cannot prove topology.

Include negative controls: intentionally remove a motif, fill a hole, swap two colours and omit one set piece in non-publishable test fixtures. The verification pipeline must reject them. Do not align outputs with unrestricted optical flow that can conceal structural drift.

Report numerator/denominator for first-pass acceptance, accepted raw vs hybrid outputs, best-of-two rate, total cost, cost/accept, p50/p95 latency and reviewer disagreement. Twelve outputs per arm are only screening. Zero failures in12 still leaves a roughly22% one-sided95% upper failure bound under independent-trial assumptions; similar products may be correlated. No universal guarantee follows.

Stage 2: deployment qualification

Test the selected workflow on all eight launch products and at least two scenes each. A deliberate new-drape challenge must be evaluated separately against withheld real-pose photos or validated geometry. Without that ground truth, mark it unverified; do not mix it into background-only success statistics. Paid continuation is conditional on the ceiling and

approval. Every published image must pass individually, regardless of aggregate benchmark performance.

Repository findings: reuse the working system

Repository: StateFarm91/Project-Money. Default HEAD inspected: 7bbbe72e205bd922297c398e0cce7e714479eeed (12 October). Newer relevant integration inspected: claude/w4-INTEG2 at 5cc096fe0d311f913c79e08838acb5c810d39182 (13 October). Not every branch tip was dated, so this is not a claim that this is the globally newest code. No live production readback, ArtifactStore inspection, test execution or actual PNG/PDF visual inspection was performed. Root CURRENT_STATE is stale and was not used as current evidence. [R1]

What the actual requirements say

The repository extraction master_v1.1.txt was read; its original PDF was located but not independently rendered. F-030 covers later gallery frames including lifestyle/use and category proof. F-254 covers hero, alternate angle, stitch detail, scale/context, construction, colour and applicable fit/worn imagery. F-253 independently requires photographic realism and exact pattern fidelity for photographic assets. Later D-FB-9/VISROLE rules retain role-specific review for disclosed informational renders. Therefore one attractive lifestyle image cannot by itself close the broader gallery requirements. F-278 concerns truthful clarity and click appeal at Etsy-like thumbnail/mobile size. Build2 #61 requires the image to communicate its role/caption claim and independent visual review. [R2-R3]

Most important confirmed code-level gap

visual/disclosed_render.py creates a 2000x2000 opaque RGB canvas and draws stitch glyphs/polygons. It is not a photographed yarn surface or transparent cutout. Converting this entire image to RGBA makes the entire canvas opaque. The existing visual/compose.py protects nonzero-alpha pixels, so blindly passing that image protects the whole background too. A separate trustworthy product source and verified matte are prerequisites to useful background editing. This finding comes from code, not visual inspection of a deployed image. [R4-R5]

visual/product_authority.py explicitly keeps structural truth UNKNOWN without qualified whole-product construction/gauge evidence, even when protected-pixel preservation passes. publish/owned_photography.make returns a redraw refusal before its old generation path. Switching a model ID cannot legitimately bypass either condition. The compositor certifies neither construction nor photorealism. [R5-R7]

What is already reusable

- gateway/images.py has provider routing, request persistence, conditioning receipts and input hashes. Its inspected table includes flux-2-pro, gpt-image-2, nano-banana-2 mapped to gemini-3.1-flash-image, and seedream-v5-lite. Embedded historical prices should be repriced, not copied into the benchmark. No Ideogram adapter was present. [R8]

- The BFL request path sends only refs[0] despite a table-level multi-reference capability. Benchmarking multiple references needs a payload assertion. A provider label is not proof that all inputs were transmitted. [R8]

- visual/contents_frame binds actual released PDF bytes and hashes. Keep this for truthful 'what is included' gallery frames. [R9]

- Newer B2VISION resolves image artifacts by digest rather than stale temporary paths. VISROLE routes chart/card reviews separately from photographs, while preserving photographic checks for photo/unknown roles. Retain these fixes. [R10-R11]

- FIRSTIMG/HEROVISION/HEROTOURNEY provide recognition at small sizes and truth-gated candidate review. The inspected notes still report insufficient truthful alternatives for some flat products. This is an evidence/candidate problem, not permission to pad a tournament. [R12]

- Spend reservation, uncertain-call accounting and bounded experiment plans exist. Reuse them; do not create a second budget subsystem. [R8,R13,R17]

Historical claims reconciled

The 'five certified galleries' evidence refers to five shadow-certified releases across basket S/M/L, Cloudline blanket and hexagon coaster families. LIFESTYLE remained uncovered. Certification is not proof of five photographic lifestyle galleries or publication. [R14]

Old one-of-six ornament and one-of-two placemat findings have subsequent MULTIPIECE/IMGFIX repair records; do not repeat them as established current defects. Similarly, an October7 funding blocker was superseded by October12 vision-probe evidence in the repository. These are historical records, not a fresh production audit. [R10,R15]

The CA$4.76 VB-1 proposal was recorded as unexecuted on October7. Code also contains a CA$100 monthly combined ceiling and a CA$50 lifetime benchmark envelope, but neither establishes current headroom or owner approval. The proposed CA$30 experiment must fit inside remaining approved limits, not add a new independent allowance. [R13,R16]

Claude Code implementation packet

Problem: Brambleloop can preserve known pixels and certify informational frames, but the inspected source renderer does not provide a qualified photographic whole-product reference. General redraws remain correctly blocked. Add a narrow testable editing adapter and reference qualification workflow, without weakening existing hard gates.

Files and changes proposed

1. Confirm owner-selected current branch and exact commit before work. Preserve active branches and frozen release candidates. This research authorizes no implementation or deployment.

2. gateway/images.py: add a dedicated Ideogram4.5 adapter, not an inherited OpenAI dialect. Endpoint POST https://api.ideogram.ai/v2/image/precise-edit/ideogram-4-5. Use image, same-size mask, explicit/auto context window, quality and optional references. Context regions are limited to4,194,304 pixels, at least256 pixels per side and aspect<=6:1; validate exact requests with the documented unbilled dry_run option before paid execution. Authenticate with an approved existing credential. No credential provisioning under this research authorization. [S2]

3. visual/render_contract.py and disclosed_render.py: characterize actual inputs; do not relabel glyph illustrations as photoreal references. Introduce a reference manifest for separately qualified photo/render sources, not a new permission to redraw products.

4. visual/compose.py: reuse existing preservation verification for its supported exact-copy mode. Design a separately named and tested alpha-edge mode only if needed, with explicit verifier qualification. Do not quietly change the old invariant or enable resampling/relighting in production.

5. visual/product_authority.py: keep UNKNOWN for missing construction/gauge evidence. Define the evidence a new qualified reference producer must supply, including pattern version, measurements, provenance, independent construction review and immutable source bytes. No 'pass if provider says consistent' shortcut.

6. visual/provider_trial.py and visual/spend_plan: adapt the existing experiment scaffold to current launch products and model versions, same-reference arms, cumulative actual costs and early-stop rules. The historical graphghan name has drifted; resolve current CIR identity instead of trusting old trial labels.

7. runtime/release._review_frames and visual/frame_role.py: keep role-specific review and independent graders. Full photo requirements remain for photographic/unknown roles. Treat a generated lifestyle scene as such even if its product comes from a deterministic source.

8. Existing gallery/listing certification: require exact final image hash, product version, qualified source and whole-gallery coverage. Preserve F-278 truthful candidate eligibility and 230/340px recognition/legibility tests.

Data contract and flow

Approved CIR + pattern/PDF hash -> independently qualified product reference -> immutable source/matte manifest -> price reservation -> provider request/conditioning receipt -> durable raw artifact -> deterministic composite -> source/product/edge/realism checks -> blind review -> final hash certificate -> listing assets -> storefront readback.

Manifest fields should include product/release ID, CIR hash, PDF hash, source type, source and matte digests, component IDs/count, reference producer/verifier version, canvas/colour space, protected region and edge policy,

approved transform or none, provider/model snapshot, prompt/parameter digest, request ID, billed usage, result digest, required disclosure and independent reviewer evidence. Reject any mismatch, missing source, uncertain price or unknown gate.

Required tests

- Same-sized source/mask bytes actually transmitted; Ideogram mask polarity black=edit differs from Recraft/Stability. Test empty/full/inverted masks, no editable area and image dimension changes.

- All references transmitted, correctly ordered and hashed; no silent first-reference truncation. Missing references and competitor-source hashes fail closed.

- Mutating one protected opaque pixel fails. Source-hash swap, obsolete product version, wrong component count, filled hole and altered border fail. Soft-edge tests use a documented baseline and catch halos.

- No generated repair is accepted merely because an embedding score is high. Deliberately corrupted negative controls must be rejected.

- Preserve current frame-role dispatch. Unknown photo checks stay UNKNOWN, not PASS. A product source with no whole-construction evidence remains UNKNOWN after perfect compositing.

- Durable artifacts survive job/container boundaries; graders do not receive expired /tmp paths. Clean all temporary files on success, exception, cancellation and timeout.

- Reserve before call; store request IDs; reconcile uncertain outcomes before retry. Count billed rejects, downloads that fail and review costs. Never multiply an uncertain paid request by blind retry.

- Verify dimensions, MIME, lossless master, derivative export, disclosure, gallery roles and thumb recognition. Two truthful hero candidates required where F-278 applies.

Reuse existing tests before adding targeted cases: test_w4_visrole, test_cert_listing_frames, test_w4_firstimg, test_w4_herovision, test_w4_herotourney, test_w4_b2vision, test_w4_multipiece, test_w4_imgfix, test_final_visual_authority, test_image_reference_hash_failclosed, test_model_spend_paths, test_w4_visual_certified_gallery, test_listing_set and the current required aggregate suite. Their names were found in repository evidence; this research did not run them.

Acceptance and effort

Deliver the reference manifest, every raw/hybrid output, blinded review, failure reasons, measured acceptance and costs, plus no-change evidence for publication gates. Do not promote the adapter or close F-030/F-254 until all relevant gallery roles and production readback pass on the intended release.

Planning only: 0.5-1 engineer-day to materialize and audit references; 2-4 days for one adapter and offline tests with reusable infrastructure; 1-2 days for an approved screening/review cycle. This may overlap and is not a delivery commitment. Missing physical/validated-render references, yarn-level modelling, owner reviews or unclear Etsy policy can dominate elapsed time. Do not promise a fixed eight-product completion date before Stage0.

Etsy suitability and truthful disclosure

Official Creativity Standards permit seller-designed digital downloads. Official image rules do not expressly resolve the ordinary digital-crochet-pattern case for auxiliary AI lifestyle renders. There is no sound basis here for either a blanket 'labelled AI mockups are allowed' claim or a blanket 'a PDF must have a manufactured product photo' claim. Obtain a specific Etsy ruling before relying on synthetic lifestyle imagery as a listing-policy solution. No support contact was made. [S36-S38]

Conservative presentation: show the actual PDF deliverable, clearly state DIGITAL CROCHET PATTERN / NO FINISHED ITEM, and use owned photographs of a genuinely made and verified sample where available. Identify any permitted synthetic scene or render accurately. AI-created-item disclosure is a separate question from marketing-image compliance; if AI created the sold pattern or included art, disclose that accurately. A label cannot cure wrong stitches, unsupported dimensions or an impossible finished result. [S36-S39]

Etsy recommends at least2000 pixels in both dimensions. A model's '2K' long-edge output may not meet that recommendation in both directions. Use an appropriate square or sufficiently large canvas and keep thumbnail-safe composition. This recommendation is not a claim that every smaller image is rejected. [S40]

Suggested provenance wording, subject to policy confirmation and the actual source:

- Real sample, synthetic setting: 'Photograph of a sample made from this pattern; background digitally generated.'

- Qualified render: 'Digital visualization of the pattern design; not a photograph of a handmade sample.'

- Unverified concept: keep internal or clearly conceptual; do not claim it proves what the pattern produces.

Do not fabricate physical testing, claim a generated scene is a genuine photoshoot, remove required provenance, or count a disclosed concept as a certified product photo.

Final recommendation

Approve preparation of the reference audit first. If usable whole-product sources exist, benchmark Ideogram4.5 against RecraftV3, Sunburst, Nano Banana2.1 and a product-background specialist under a single bounded spend plan. Keep FLUX3, Gemini3Pro, Seedream5 and Qwen3 as serious challengers when a particular failure justifies another arm.

The best available solution is likely not a wholesale architecture rewrite. Brambleloop already owns much of the preservation, artifact and gate infrastructure. The highest-leverage missing piece is trustworthy photographic-looking product evidence, followed by a controlled background-edit adapter. A newer model may solve much of the visual quality problem; only the controlled test can establish whether it does so for these eight actual designs.

Source register

Official sources accessed 19 October2026. URLs are clickable. Documentation can change; pin model snapshots and recheck prices/contracts before an approved trial.

S1 Bank of Canada daily FX: https://www.bankofcanada.ca/rates/exchange/daily-exchange-rates/

S2 Ideogram Precise Edit API: https://developer.ideogram.ai/api-reference/images/precise-edit/ideogram-4-5

S3 Ideogram pricing: https://ideogram.ai/pricing/?tool=ideogram

S4 Ideogram4.5 generation: https://developer.ideogram.ai/api-reference/images/generate/ideogram-4-5

S5 Recraft inpainting: https://www.recraft.ai/docs/api-reference/tools/inpainting

S6 Recraft background tools and price: https://www.recraft.ai/docs/api-reference/tools/background ; https://www.recraft.ai/docs/api-reference/pricing

S7 Recraft developer terms: https://www.recraft.ai/legal/developer-terms

S8 OpenAI Sunburst model: https://developers.openai.com/api/docs/models/gpt-image-2.5-sunburst

S9 OpenAI image guide: https://developers.openai.com/api/docs/guides/image-generation

S10 OpenAI pricing: https://developers.openai.com/api/docs/pricing

S11 OpenAI dated releases and edits API: https://developers.openai.com/api/docs/changelog ; https://developers.openai.com/api/reference/resources/images/methods/edit

S12 Google current model: https://ai.google.dev/gemini-api/docs/models/gemini-nano-banana-2.1 ; https://ai.google.dev/gemini-api/docs/models/gemini-3-pro-image

S13 Google image generation: https://ai.google.dev/gemini-api/docs/image-generation

S14 Google pricing and GA specifications: https://ai.google.dev/gemini-api/docs/pricing ; https://docs.cloud.google.com/gemini-enterprise-agent-platform/models/gemini/nano-banana-2-1

S15 BFL3 API: https://docs.bfl.ai/api-reference/utility/generate-an-image-with-flux-3

S16 BFL3 overview: https://docs.bfl.ai/flux_3/flux3_image_overview

S17 BFL pricing: https://docs.bfl.ai/quick_start/pricing ; https://bfl.ai/pricing?category=flux.2

S18 Adobe current API: https://developer.adobe.com/firefly-services/docs/firefly-api/

S19 Adobe precise composition and pricing: https://developer.adobe.com/firefly-services/docs/firefly-api/guides/how-tos/object-composite/ ; https://business.adobe.com/products/firefly-business/firefly-services.html

S20 Photoroom controls and price: https://docs.photoroom.com/image-editing-api-plus-plan/ai-backgrounds ; https://docs.photoroom.com/image-editing-api-plus-plan/keep-existing-transparency ; https://www.photoroom.com/api/pricing

S21 Bria lifestyle and price: https://docs.bria.ai/product-shot-editing/products/product-lifestyle-shot-by-text ; https://bria.ai/pricing

S22 Seedream5 models, price, terms: https://docs.byteplus.com/en/docs/ModelArk/seedream-5-0-pro ; https://docs.byteplus.com/id/docs/modelark/model-pricing ; https://docs.byteplus.com/en/docs/legal/docs-service-specific-terms

S23 Qwen hosted API and price: https://www.alibabacloud.com/help/en/model-studio/qwen-image-generation-and-editing-api-reference ; https://www.alibabacloud.com/help/en/model-studio/model-pricing

S24 Qwen open editing: https://huggingface.co/Qwen/Qwen-Image-Edit-2511

S25 Runway API and prices: https://docs.dev.runwayml.com/guides/using-the-api/ ; https://docs.dev.runwayml.com/guides/pricing/

S26 Runway references and commercial rights: https://help.runwayml.com/hc/en-us/articles/40042718905875-Creating-with-Gen-4-Image-References ; https://dev.runwayml.com/learn/commercial-use

S27 Stability pricing, API and license: https://platform.stability.ai/pricing ; https://platform.stability.ai/docs/api-reference ; https://stability.ai/license

S28 Stability hosted terms: https://platform.stability.ai/legal/terms-of-service

S29 FLUX open weights: https://huggingface.co/black-forest-labs/FLUX.2-klein-4B ; https://huggingface.co/black-forest-labs/FLUX.2-dev

S30 Tencent models and license: https://github.com/Tencent-Hunyuan/HunyuanImage-3.0 ; https://github.com/Tencent-Hunyuan/HunyuanImage-3.0/blob/main/LICENSE

S31 Tencent background API and Flair: https://cloud.tencent.com/document/product/1668/107957 ; https://flair.ai/key-features/flair-api ; https://flair.ai/pricing

S32 OpenAI Services Agreement: https://openai.com/policies/services-agreement/

S33 Gemini API terms: https://ai.google.dev/gemini-api/terms

S34 Ideogram legal/API terms: https://ideogram.ai/legal/tos/ ; https://ideogram.ai/legal/api-tos/

S35 BFL commercial/data terms: https://bfl.ai/legal/developer-terms-of-service ; https://bfl.ai/legal/flux-api-service-terms

S36 Etsy image policy: https://www.etsy.com/legal/policy/listing-image-requirements/253962679005

S37 Etsy Creativity Standards: https://www.etsy.com/legal/creativity

S38 Etsy Seller Policy: https://www.etsy.com/legal/sellers/

S39 Etsy AI explanation: https://www.etsy.com/seller-handbook/article/1275449912004

S40 Etsy image requirements: https://help.etsy.com/hc/en-us/articles/115015663347-Requirements-and-Best-Practices-for-Images-in-Your-Etsy-Shop

Repository links below use inspected integration commit5cc096fe0d311f913c79e08838acb5c810d39182, not a moving branch.

R1 Inspected integration: https://github.com/StateFarm91/Project-Money/tree/5cc096fe0d311f913c79e08838acb5c810d39182

R2 brambleloop/research/final_build/master_v1.1.txt: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/research/final_build/master_v1.1.txt

R3 brambleloop/src/brambleloop/build2/requirements.json: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/src/brambleloop/build2/requirements.json

R4 brambleloop/src/brambleloop/visual/disclosed_render.py: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/src/brambleloop/visual/disclosed_render.py

R5 brambleloop/src/brambleloop/visual/compose.py: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/src/brambleloop/visual/compose.py

R6 brambleloop/src/brambleloop/visual/product_authority.py: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/src/brambleloop/visual/product_authority.py

R7 brambleloop/src/brambleloop/publish/owned_photography.py: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/src/brambleloop/publish/owned_photography.py

R8 brambleloop/src/brambleloop/gateway/images.py: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/src/brambleloop/gateway/images.py

R9 brambleloop/src/brambleloop/visual/contents_frame.py: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/src/brambleloop/visual/contents_frame.py

R10 brambleloop/research/final_build/w4/B2VISION.md: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/research/final_build/w4/B2VISION.md

R11 brambleloop/research/final_build/w4/VISROLE.md: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/research/final_build/w4/VISROLE.md

R12 brambleloop/research/final_build/w4/HEROTOURNEY.md: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/research/final_build/w4/HEROTOURNEY.md

R13 brambleloop/src/brambleloop/finance/spend_policy.py: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/src/brambleloop/finance/spend_policy.py

R14 brambleloop/research/final_build/w4/VISUALSEO_FIX.md: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/research/final_build/w4/VISUALSEO_FIX.md

R15 brambleloop/research/final_build/w4/IMGFIX.md: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/research/final_build/w4/IMGFIX.md

R16 brambleloop/research/final_build/w4/VISUAL_STATUS.md: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/research/final_build/w4/VISUAL_STATUS.md

R17 Bounded visual spend plans: https://github.com/StateFarm91/Project-Money/blob/5cc096fe0d311f913c79e08838acb5c810d39182/brambleloop/src/brambleloop/visual/spend_plan.py